TAMPINES JUNIOR COLLEGE

Preliminary Examinations 2008

CHEMISTRY (Higher 2) 9746/1

PAPER 1 Multiple Choice

Wednesday **27 August 2008** 1 hour

Additional materials: Multiple Choice Answer Sheet
Data Booklet

INSTRUCTIONS TO CANDIDATES

Write in soft pencil.
Write your name and Civics Group on the Answer Sheet in the spaces provided.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**.
Choose the one you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **10** printed pages.

[Turn over

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 Air contains 20% oxygen by volume. What volume of air is required for the complete combustion of 1.0 dm^3 of octane vapour in the car engine?

A 2.5 dm^3 **B** 12.5 dm^3 **C** 62.5 dm^3 **D** 125.0 dm^3

2 Which of the following concerning $^{79}_{34}Se$ and $^{79}_{35}Br$ is true?

A $^{79}_{34}Se$ is smaller than $^{79}_{35}Br$.

B $^{79}_{34}Se$ has more neutrons per atom than $^{79}_{35}Br$.

C $^{79}_{34}Se$ has more protons per atom than $^{79}_{35}Br$.

D Both $^{79}_{34}Se$ and $^{79}_{35}Br$ contain the same number of electrons per atom.

3 25.0 cm^3 of a solution of bromine water is treated with excess potassium iodide and the resulting iodine reacts exactly with 25.00 cm^3 of 0.01 mol dm^{-3} sodium thiosulphate solution:

$$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

What is the concentration of the bromine solution in mol dm^{-3}?

A 1×10^{-3} **B** 2×10^{-3} **C** 5×10^{-3} **D** 1×10^{-2}

4 For the pairs of species shown below, in which pair does the first species have a bigger bond angle than the second?

A PH_3 and NH_3

B $SOCl_2$ and OCl_2

C SO_3^{2-} and CO_3^{2-}

D BrF_2^- and $BeCl_2$

5 Which one of the following sets of compounds consists of giant ionic structure, a giant covalent structure and a simple covalent structure?

A $C_6H_5CO_2H$, P_4O_{10} and SiC

B BaO_2, ICl_3 and H_3PO_4

C $BeCl_2$, $BaCl_2$ and SiO_2

D C (diamond), $AlCl_3$ and $BeCl_2$

6 In a syringe experiment, 0.10 g of a gas is found to occupy 83.1 cm^3, measured at standard pressure (1.0 x 10^5 Pa) and 27°C.
What is the relative molecular mass of the gas?

A $\dfrac{0.1 \times 8.31 \times 27}{1.0 \times 10^5 \times 83.1}$

B $\dfrac{0.1 \times 8.31 \times 27}{1.0 \times 10^5 \times 83.1 \times 10^{-6}}$

C $\dfrac{0.1 \times 8.31 \times 300}{1.0 \times 10^5 \times 83.1}$

D $\dfrac{0.1 \times 8.31 \times 300}{1.0 \times 10^5 \times 83.1 \times 10^{-6}}$

7 The major natural source of fluorine is the mineral fluorspar, which is mainly calcium fluoride, CaF_2. The first stage in liberating F_2 is to grind the compound up and react it with concentrated sulphuric acid. The products are hydrogen fluoride and calcium sulphate.

$$CaF_2 + H_2SO_4 \rightarrow 2HF + CaSO_4$$

What is the enthalpy change of the above reaction (in kJ mol^{-1}), given the following information?

compound	**ΔH_f (kJ mol^{-1})**
CaF_2	–1220
H_2SO_4	–814
HF	–271
$CaSO_4$	–1434

A +58 **B** –58 **C** +329 **D** –329

8 Which of the following is an effect of a catalyst on an endothermic and reversible reaction?

A The reaction becomes exothermic.
B The equilibrium position of the reaction shifts to the right.
C The reaction proceeds at a faster rate.
D Stoichiometry of the reaction is affected.

9 Compound P decomposes on heating according to the following equation:

$$2P(g) \rightleftharpoons Q(g) + 2R(g)$$

When 5 mol of P were put into a 1 dm^3 container and heated, the equilibrium mixture contained 0.6 mol of Q.
What is the numerical value of the equilibrium constant, K_c?

A $\dfrac{(0.6)^2 \times 0.3}{5^2}$

B $\dfrac{(0.6)^2 \times 1.2}{5^2}$

C $\dfrac{0.6 \times (0.3)^2}{3.8^2}$

D $\dfrac{0.6 \times (1.2)^2}{3.8^2}$

10 Consider the following equilibrium reaction in a closed vessel:

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

At 800°C, the equilibrium constant is 0.236 atm.

Which one of the following statements is **TRUE** about the equilibrium?

A The position of the equilibrium shifts to the right when temperature is decreased.
B The position of the equilibrium shifts to the right when pressure is decreased.
C The value of the equilibrium constant decreases with increasing temperature.
D The value of the equilibrium constant decreases when less $CaCO_3$ is used.

11 A solution contains 1×10^{-3} mol dm^{-3} of bromide, fluoride, iodide and sulphate ions. Which of the following lead (II) compounds will be precipitated first when 0.01 mol dm^{-3} of lead (II) nitrate is added dropwise into the solution?
[*The solubility products of the compounds are given.*]

	compound	K_{sp} *(at 25°C)*
A	lead (II) bromide	4.0×10^{-5}
B	lead (II) fluoride	2.7×10^{-8}
C	lead (II) iodide	7.1×10^{-9}
D	lead (II) sulphate	1.6×10^{-8}

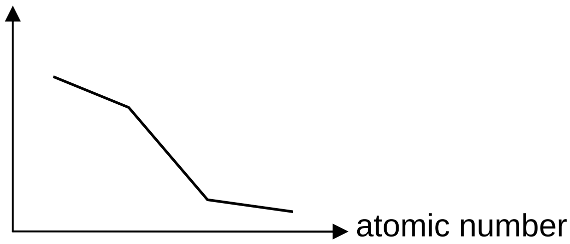
12 What is the pH of an aqueous solution containing 0.1 mol dm^{-3} sodium benzoate and 0.01 mol dm^{-3} benzoic acid? [K_a of benzoic acid = 6×10^{-5} mol dm^{-3}]

A 3.22 **B** 4.22 **C** 4.78 **D** 5.22

13 A current of 8 A is passed for 100 minutes through dilute aqueous copper sulphate using inert electrodes. What will be the approximate volume of gas liberated measured at s.t.p.?

A 2.8 dm^3 **B** 5.6 dm^3 **C** 8.4 dm^3 **D** 11.2 dm^3

14 Which property when plotted against increasing atomic number gives the shape of the following graph?



A ionic radius of Mg, A*l*, Si and P
B first ionisation energy of Mg, Ca, Sr and Ba
C boiling point of chlorides of Na, Mg, Al and Si
D electronegativity of Si, P, S and C*l*.

15 "*Group II metals have higher melting points than Group I metals.*"

A student suggested four possible reasons to explain the above statement. Which of the following statements is most likely to be correct?

A There is a larger overlap of valence electrons between Group II metal atoms than between Group I metal atoms.
B Two valence electrons per atom for Group II metals are available for bonding in the metallic lattice as compared to one valence electron per atom for Group I.
C Group II metals have higher total ionization energies than Group I metals during formation of ions with octet electronic configuration.
D Group II metal atoms have larger atomic radius than Group I metal atoms.

16 A mixture of iodine and chlorine was reacted with excess sodium thiosulphate solution. The resulting solution was treated with excess aqueous barium nitrate. Precipitate **T** formed was filtered off and washed with distilled water.
The filtrate was then treated with silver nitrate followed by excess aqueous ammonia. The mixture was again filtered leaving behind precipitate **U**. The resultant filtrate was treated with aqueous lead nitrate to give a white precipitate **V.**

Which anion was present in each of the precipitates respectively?

	Precipitate **T**	Precipitate **U**	Precipitate **V**
A	Cl^-	I^-	NO_3^-
B	I^-	SO_4^{2-}	NO_3^-
C	SO_4^{2-}	I^-	Cl^-
D	SO_4^{2-}	NO_3^-	Cl^-

17 Which of the following statements is TRUE for the Group VII hydrides, HC*l*, HBr and HI?

A Enthalpy change of formation becomes less exothermic from HC*l* to HI.
B Thermal stability increases from HC*l* to HI.
C HI cannot be oxidised to iodine by any of the other halogens.
D An aqueous solution of HC*l* is the strongest acid.

18 The following species all have either a half- or fully-filled 3d subshell except

A Cr
B Zn^{2+}
C V
D Cu^+

19 Which of the following shows the correct order of increasing pK_a for the compounds given below?

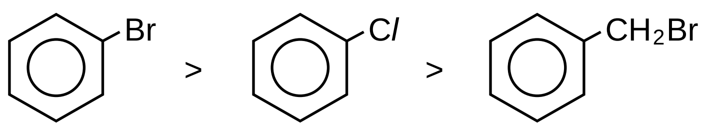
$\boxed{CH_3CH_2COOH,\ ClCH_2CH_2COOH,\ CH_3CH_2CH_2OH,\ CH_3CH_2CONH_2}$

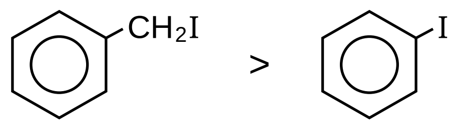
A $ClCH_2CH_2COOH < CH_3CH_2COOH < CH_3CH_2CH_2OH < CH_3CH_2CONH_2$
B $CH_3CH_2COOH < ClCH_2CH_2COOH < CH_3CH_2CH_2OH < CH_3CH_2CONH_2$
C $CH_3CH_2CONH_2 < CH_3CH_2CH_2OH < CH_3CH_2COOH < ClCH_2CH_2COOH$
D $CH_3CH_2CONH_2 < CH_3CH_2COOH < CH_3CH_2CH_2OH < ClCH_2CH_2COOH$

20 Which sequence shows the correct order of decreasing ease of hydrolysis?

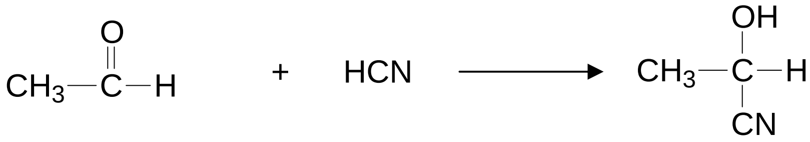
A $CH_3CH_2COCl > CH_3CH_2Cl > CH_2{=}CHCl$

B $CH_3CH_2Cl > CH_3CH_2Br > CH_3CH_2I$

C 

D $CH_2{=}CHI$ > 

21 Ethanal reacts with CN^- from HCN in the presence of a weak base as shown below:



In a similar reaction, $^-CH_2COCH_3$ ions are generated when CH_3COCH_3 reacts with a strong base. Which one of the following compound is the product when ethanal reacts with $^-CH_2COCH_3$?

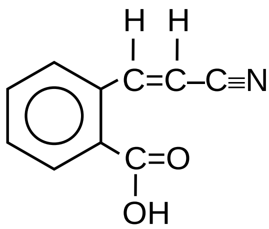
A $CH_3CH(OH)CH_2COCH_3$

B $(CH_3)_2C(OH)CH_2CHO$

C $(CH_3)_2C(CHO)CH_2OH$

D $(CH_3)_2C(OH)CH_2COCH_3$

22 In the presence of nickel, how many moles of H_2 react with one mole of the following compound?



A 2 **B** 3 **C** 4 **D** 5

23 2,2–dimethylpropylamine can be produced by this 2-step reaction scheme starting with compound **E**.

$$\mathbf{E} \xrightarrow[\text{reflux}]{\text{KCN (ethanol)}} \xrightarrow[\text{Pt catalyst}]{H_2(g)} CH_3C(CH_3)_2CH_2NH_2$$

2,2–dimethylpropylamine

Which one of the following structures represents **E?**

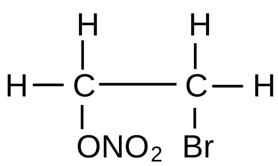
A $CH_3C(Br)(CH_3)_2$

B $CH_3(CH_2)_2CH_2Br$

C $BrCH_2CH(CH_3)_2$

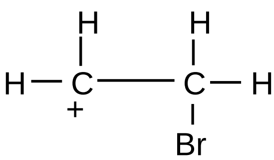
D $CH_3CH(CH_2Br)CH_3$

24 When ethene reacts with bromine in the presence of concentrated aqueous sodium nitrate, the product contains the following compound.

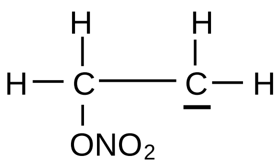


What is the intermediate formed in this reaction?

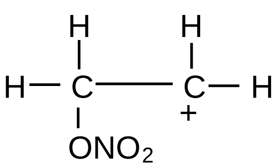
A



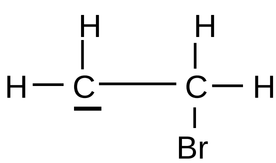
C



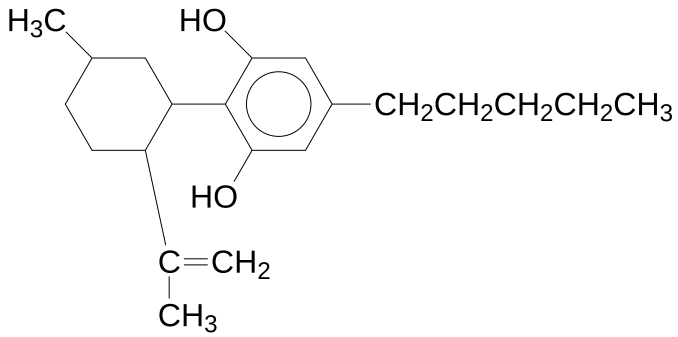
B



D



25 The psychoactive drug in cannabis has the following structure:



When it is completely reacted with HBr, how many chiral centres does a molecule of the **major** product possess?

A 1 **B** 3 **C** 4 **D** 6

26 A compound **R**, with molecular formula $C_9H_{10}O_2$, is heated under reflux with aqueous sodium hydroxide and the resulting mixture then cooled and acidified with dilute sulphuric acid. The final products include a compound that turns blue litmus solution red, and another which gives a purple colouration when tested with aqueous iron(III) chloride. What could **R** be?

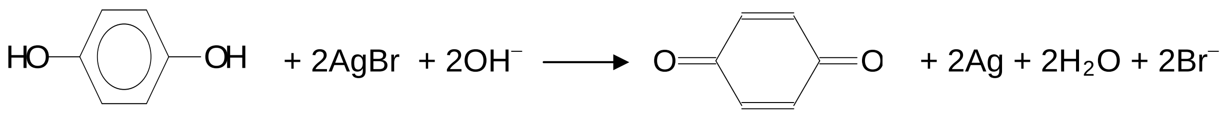
A $C_6H_5OCOCH_2CH_3$

B $C_6H_5CH_2OCOCH_3$

C $C_6H_5OCH_2COCH_3$

D $C_6H_5CH_2CH_2COOH$

27 In developing exposed film from a camera, the light-activated sliver bromide crystals is reacted with aqueous alkaline hydroquinone as follows:

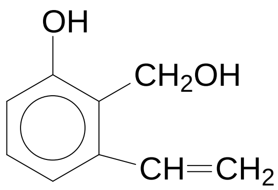


hydroquinone

Which one of the following best describes the role of hydroquinone?

A an acid only
B an oxidising agent only
C a reducing agent only
D both an acid and reducing agent

28 A compound has the following structure:



All of the following reagents will react with the above compound, except

A aqueous bromine
B phosphorous pentachloride
C hydrogen chloride
D potassium cyanide

29 A compound **P** reacts with PCl_5 to form hydrogen chloride gas and decolourises hot acidified potassium manganate(VII). Which of the following could **P** be?

A HOOC–COOH

B CH_3CHO

C CH_3COCH_3

D $(CH_3)_3COH$

30 A sample of aminoethanoic acid is dissolved in a buffer solution of pH 4. Which of the following gives the two main forms of aminoethanoic acid at this pH?

A $^{+}NH_3CH_2CO_2H$ and $^{+}NH_3CH_2CO_2^{-}$
B $^{+}NH_3CH_2CO_2H$ and $NH_2CH_2CO_2H$
C $^{+}NH_3CH_2CO_2H$ and $NH_2CH_2CO_2^{-}$
D $^{+}NH_3CH_2CO_2^{-}$ and $NH_2CH_2CO_2^{-}$

Section B

For each of the questions in this section one or more of the three numbered statements **1** *to* **3** *may be correct.*

Decide whether each of the statements is or not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** *to* **D** *should be selected on the basis of*

A	B	C	D
1,**2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

31 In which of the following reactions does the oxidation number of nitrogen change by two?

1 $NH_2OH \rightarrow NH_3$
2 $N_2 \rightarrow NO$
3 $NO_2 \rightarrow HNO_3$

32 Which one of the following solids contains atoms held together by covalent bonds?

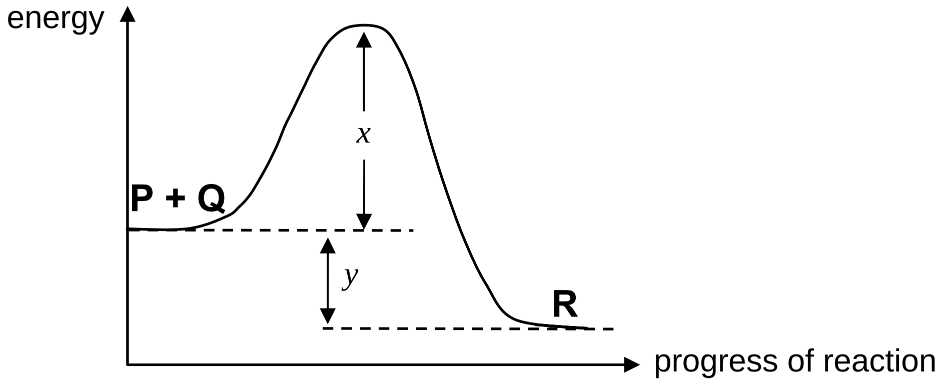
1 $H_2O(s)$ **2** $CO_2(s)$ **3** $SiO_2(s)$

33 In the ideal gas Law equation: $PV = nRT$, each symbol has its usual meaning. Which of the following statements are correct?

1 Real gas deviates most from ideal gas behaviour at high pressure and low temperature.
2 The density of an ideal gas at constant pressure is inversely proportional to the absolute temperature.
3 The volume of the gas of an ideal gas is doubled when its temperature is raised from 25°C to 50°C at constant pressure.

A	B	C	D
1,**2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

34 The energy profile diagram for a reversible reaction **P** + **Q** $\rightleftharpoons$ **R** is shown below.



Which of the following statements are correct?

1 The activation energy of the backward reaction is $x + y$.
2 The magnitude of the enthalpy change of the backward reaction is $x - y$.
3 The forward reaction is endothermic.

35 Some standard redox potentials values are given in the table below.

Electrode reaction	$E^\ominus$/ V
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	+0.77
$Sn^{4+}(aq) + 2e^- \rightleftharpoons Sn^{2+}(aq)$	+0.15
$Ce^{4+}(aq) + e^- \rightleftharpoons Ce^{3+}(aq)$	+1.45

Which of the following reactions would be expected to occur under standard conditions?

1 $Ce^{4+}(aq) + Fe^{2+}(aq) \rightarrow Ce^{3+}(aq) + Fe^{3+}(aq)$
2 $Sn^{2+}(aq) + 2Fe^{3+}(aq) \rightarrow Sn^{4+}(aq) + 2Fe^{2+}(aq)$
3 $2Ce^{4+}(aq) + Sn^{2+}(aq) \rightarrow 2Ce^{3+}(aq) + Sn^{4+}(aq)$

36 The hexa-aquairon(III) ion hydrolyses as shown below.

$$[Fe(H_2O)_6]^{3+}(aq) + H_2O(l) \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+}(aq) + H_3O^+(aq)$$

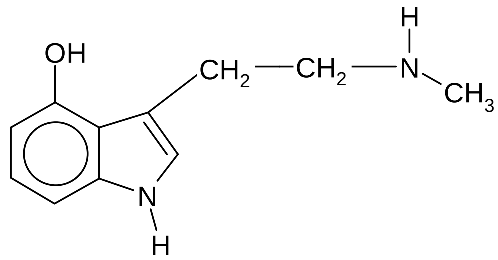
Which of the following statements about the above equilibrium are correct?

1 The hydrolysis is favoured by low pH values.
2 The iron does not undergo a change in oxidation state.
3 The corresponding iron(II) in $[Fe(H_2O)_6]^{2+}$ is less likely to undergo hydrolysis.

37 For the sequence hydrogen chloride, hydrogen bromide and hydrogen iodide, there is

1 a decrease in thermal stability.
2 an increase in bond length.
3 an increase in ease of oxidation.

38 *Psilocin* is a psychedelic mushroom alkaloid. It is the active compound that produces hallucinations from ingesting "magic mushrooms" and amplifies sensory experience. Compound **X** is a derivative of *Psilocin*.

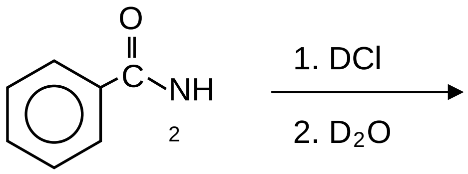


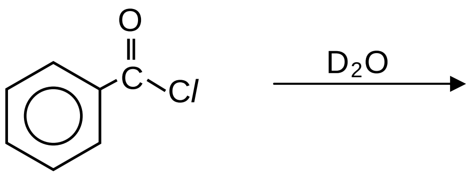
X

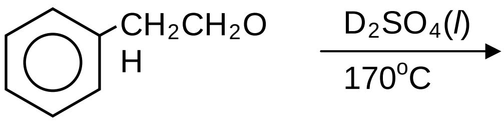
Which of the following statements are true for compound **X**?

1 It gives white fumes with CH_3COCl.
2 It dissolves in both aqueous acids and alkalis.
3 It is hydrolysed by dilute acids.

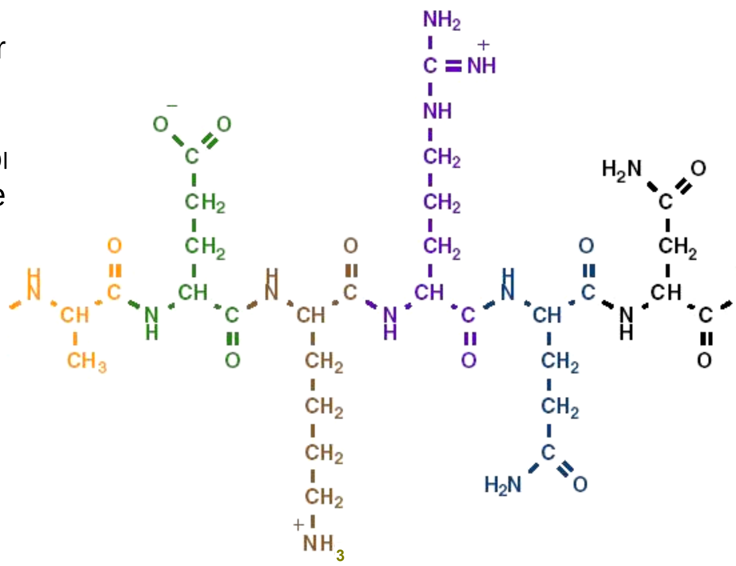
39 Which of the following will yield an organic compound containing deuterium, D? (D = ^{2}H)

1 

2 

3 

40 The structure on the right shows par
a protein.
Which of the following types of
interactions would be responsible fo
maintenance of the tertiary structure
the protein?



1 ionic interactions
2 hydrogen bonding
3 van der Waals' forces

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